Modeling Pursuit-Evasion Games with Numerical Simulation of Stochastic Differential Equations

Overview

We explore pursuit-evasion games as a multi-dimensional extension to the theory of Brownian bridges. Applications for this topic include predator-prey interactions, search-and-rescue scenarios, and strategies for competitive sports. We aim to study the dynamics of evasion and pursuit strategies, as well as gain insight into decision-making in uncertain environments and interactions.

Background and Motivation

Pursuit-evasion games encompass a class of problems and their corresponding mathematical frameworks, where two teams—the pursuers and the evaders— have an adversarial relationship (Weintraub et al., 2020). The classic pursuit-evasion game, developed by Isaacs, consists of one pursuer and one evader, where the pursuer's goal is to maximize the probability of capturing the evader, while the evader's goal is to minimize the probability of being captured; however, in the original formulation, the pursuers and the evaders have perfect knowledge of each other's states. In the sections below, we provide two examples of such a problem.

Example 1

Consider a pursuer named Alice and a prey named Bob in a maze environment (which can range from one up to multiple dimensions). Alice will have stochastic drift in the direction of the evading Bob who will attempt to escape before time runs out and he is caught. In particular, we can add the more realistic assumption that the predator cannot change directions instantaneously and will have an exponentially-weighted decay on momentum (or alternatively, as a function of the prey's velocity).

Example 2

The classic example of the pursuit-evasion problem is the homicidal chauffeur problem (Isaacs, 1951). As shown in **Figure 1**, there is a homicidal chauffeur (red) chasing after a pedestrian (blue) in an unbounded space. The pedestrian moves slower than the vehicle and has a maximum speed, but can change directions at will; the car has a minimum turn radius when changing directions (illustrated by the dotted lines). The game assumes that both players are aware of each other's exact position, speeds, capture radius, and turn radius (Merz, 1971). The chauffeur aims to minimize the capture time, while the person aims to maximize that time; capture is defined as when the pedestrian's position intersects with the car's boundary.

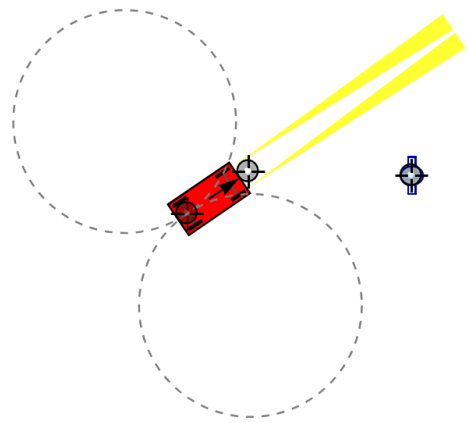

Figure 1. Visualization of the Homicidal Chauffeur problem (Becker and Garcia, 2018)

The car's movement is modeled as follows

$$\frac{d}{dt}x = v_{car} \cos\theta$$

$$\frac{d}{dt}y = v_{car} \sin\theta$$

$$\frac{d}{dt}\theta = v_{car} u$$

where r_{min} is the minimum turning radius, and u is a value between $-1/r_{min}$ and $1/r_{min}$ (Becker and Garcia, 2018).

The following equations represent the pedestrian's movement:

$$\frac{d}{dt}x = v_x$$

$$\frac{d}{dt}y = v_y,$$

where we must satisfy the constraint that $|(v_x, v_y)| \leq v_{car}$ (Becker and Garcia, 2018). Merz approached this problem by dividing the parameter space into 20 regions and categorizing different optimal paths and singular lines in each of the regions (Patsko and Turova, 2011).

Application

Some extensions to the classical problems include adding noise to the measurements of the adversary's position and velocity, including multiple agents, and considering a pursuit in three-dimensional space (Weintraub et al., 2020). Although many of these may seem like toy examples or trivial games, finding solutions can be quite difficult, and these problems serve as proxies for tackling important problems in a wide array of applications, including ecological behaviors, combat operations, competitive sports, and navigation.

This project focuses on a biological extension of the classic pursuit-evasion model, where both predators and prey can potentially reproduce when close to their respective parties and create new actors in the environment. The inspiration behind our extension was to simulate the fight between the predatory mites (that one of the authors reluctantly released on their prized plants as part of an integrated pest management system) and plant pests. The predatory mites leverage chemical cues to chase down and locate nearby prey, which serves as the inspiration for one of the strategies we explore. Overall, the concepts and simulations we explore can apply broadly to ecosystem and physiological modeling (Gal and Casas, 2014).

Previous work in the realm of SDEs and integrated pest management focuses on developing models to aid in the design of tactics to control pest populations (Tan et al., 2019). Many of these models are formulated as stochastic population models (extensions of the Lotka–Volterra model), rather than pursuit-evasion games. For example, predator-prey interaction for integrated pest management was formulated as follows

$$\begin{aligned} \mathrm{d}x(t) &= \left[rx(t)\left(1 - \frac{x(t)}{K}\right) - \frac{x(t)y(t)}{a + x(t)^2} \right] \mathrm{d}t + \alpha_1 x(t) \mathrm{d}B_1(t), \\ \mathrm{d}y(t) &= \left[\frac{\mu x(t)y(t)}{a + x(t)^2} - Dy(t) \right] \mathrm{d}t + \alpha_2 y(t) \mathrm{d}B_2(t), \end{aligned},$$

where x(t) and y(t) are prey and pest population densities; r is the growth rate for prey; K represents the carrying capacity for prey. μ is the consumption rate of prey, and D represents the death rate for predators (Xiao and Ruan, 2001; Huang et al., 2019). Lastly, α_i represents the intensity of the noise and can be a function. B_i is, of course, Brownian motion.

While these models have proven useful, using a pursuit-evasion perspective becomes important when considering complex environments and in cases where the habitat's architecture plays a role in population dynamics and behavior. For instance, Morice et al. observed how the environment's complexity affects the ability and behavior of the wolf spiders to perceive and pursue wood crickets; they also noted that the lack of work in studying both predator and prey strategies in a pursuit-evasion formulation has likely hindered our understanding of population dynamics and co-evolutionary relationships. In addition to ecological applications relating to predator-prey interactions, pursuit-evasion models also apply to model immune cells seeking out invaders or cancerous cells (Gal and Casas, 2014), but unfortunately, there is a dearth of literature studying these interactions in this context as well.

In this paper, we will explore a predator-prey pursuit on a 1D line and a 2D arena in which actors do not know the exact location of their adversaries. Two strategies for predators and prey will be simulated and analyzed. We will also examine cases where there are differing population counts of predators and prey and allow reproduction within the respective groups.

SDEs and Pursuit-Evasion Modeling

In this section, we describe the stochastic differential equations (SDEs) that we leverage to model the movement of the predators and prey in our simulations.

Mathematical model for the case of one prey & one predator

Let $\mathbf{X}(\omega, t)$ denote the prey's position and $\mathbf{Y}(\omega, t)$ be the predator's position in R^n. Then we can model their movement as:

$$d\mathbf{X}(\omega, t) = \mathbf{u}(\mathbf{d}(\mathbf{X}, \mathbf{Y}))dt + \mathbf{f}(\omega, t)d\mathbf{B}_t$$

$$d\mathbf{Y}(\omega, t) = \mathbf{v}(\mathbf{d}(\mathbf{X}, \mathbf{Y}))dt + \mathbf{g}(\omega, t)d\mathbf{B}_t$$

where $\mathbf{d}(,)$ is the distance metric, $\mathbf{u}$ and $\mathbf{v}$ are non-negative functions representing escape and pursuit velocity, $\mathbf{f}$ and $\mathbf{g}$ represent the erraticness of the prey and predator's movements, respectively. The Brownian motion $d\mathbf{B}_t$ is also n-dimensional. For example, a prey with better visibility than the predator may have $E[|\mathbf{f}|] < E[|\mathbf{g}|]$ so that its get-away can be more efficient than a predator in the chase.

A basic stopping condition is $0 \leq \mathbf{d}(\mathbf{X},\mathbf{Y}) < \varepsilon$, where ε is the arm's reach of the predator. However, we can also replace this threshold with a random variable $\mathbf{A}$ whose CDF $F_{\mathbf{A}}$ is a

function of the distance between the predator and prey; for instance, we can introduce a Poisson process, whose exponential rate λ is inversely proportional to d(**X**, **Y**) and the first occurrence being the capture of the prey by the predator.

Analysis of pursuit-evasion in the case of one prey and one predator

We can express the expected distance between the prey and the predator as a function of time elapsed (T > 0) since the start of the experiment until the prey is caught by the predator:

$$\mathbf{E}[|\mathbf{X_T} - \mathbf{Y_T}|] = \mathbf{d}(\mathbf{X}_0, \mathbf{Y}_0) + \int \mathbf{u}(\mathbf{d}(\mathbf{X}_t, \mathbf{Y}_t)) - \mathbf{v}(\mathbf{d}(\mathbf{X}_t, \mathbf{Y}_t))dt$$

where the terms involving Brownian motion vanish by linearity of expectation and the expected value of martingale, and noting that the integral is evaluated from t = 0 to t=T.

Mathematical model for case of multiple prey & predators

In place of the original Itô processes, **X** and **Y** are m_X x n and m_Y x n-dimensional, respectively, while **u** and **v** are m_Y x n and m_X x n-dimensional functions applied to the vector of distances from each prey (predator) to each predator (prey). Then **f** and **g** are also m_X x n x n and m_Y x n x n-dimensional.

Simulation Approach

Experiment setup

Here we outline the initial and stopping conditions of the numerical simulation which is followed by graphical analysis for problems such as hitting times in the multidimensional case that are analytically intractable.

The initial condition involves a configured number of actors (predators and prey) randomly placed with coordinates that are Gaussian distributed in n-dimensional Euclidean space. They are enclosed within a boundary where they may eat and reproduce. The actors can have strategies for survival that determine the functions **u**,**v**, **f**, **g** as introduced in our mathematical model. We run a fixed number of experimental trials, and each trial runs for a maximum number of steps but can terminate early if all predators or all prey have died (in which case there is no purpose in continuing the simulation, as the ecosystem has collapsed).

Strategies for predators and prey

There are three strategies we evaluate for the survival of the two actor species. The predators can have one strategy, and the prey can use the same (related) or a different strategy:

1. First, the most naive strategy puts $\mathbf{u} = \mathbf{v} = \mathbf{0}$, so there is no homing pursuit or escape, and we are just dealing with classical Brownian motion. Naturally, we expect this to be the weakest strategy for both species.
2. In the second strategy, each predator chases the closest prey, and each prey evades the closest predator. This is equivalent to a maximin optimization of survival time, or using the L-infinity norm, alternatively viewed as a greedy AI strategy.
3. The third idea is for predators to pursue the centroid of all remaining prey, and the prey to escape from the centroid of all living predators. The linear transformation is a unit matrix, and this minimizes (for predators) or maximizes (for prey) the expected distance, but it does not actually guarantee capture and escape in practice because the centroid is not an actual actor, merely an imaginary concept as part of a mental framework.

Modeling Capture and Escape

Since we numerically approximate continuous paths in R^n, it is not straightforward to determine whether a predator and prey have the exact same position at some point over a time interval. One proposed solution is to assume that the predator is able to attack and consume any prey within a fixed radius (i.e. distance of 1) from itself. However, there are a few drawbacks to this idea: first, the decision outcome is binary and does not model real-world interactions well (such as when a prey can get trapped in unpredictable terrain); second, if we naively distribute predators and prey at the start of the experiment, there is a chance that some prey will automatically be caught by a predator automatically which defeats the purpose of running the stochastic trial.

Instead, we can model a "fuzzy" boundary of pursuit-evasion result by simulating the event of capture or escape in each unit step (δt) by drawing from a random distribution whose probability density decreases rapidly with the distance, i.e. we can pick the exponential distribution with rate inversely proportional to the separation between a predator and prey. This way, the closer a predator is to its prey, the more likely it will catch its prey. If the sampled value is below a certain threshold, then we interpret the predator to have captured its prey. And thus,

we have resolved both issues argued earlier, as we have a continuous random variable whose distribution does not present any jumps.

Bounding box environment

In order that the prey does not run off indefinitely far away and disperse too far from reach, we have a rectangular boundary such that if the prey's position exceeds the left (or right, top, bottom) boundary, it will be placed back at approximately the left (or right, top, bottom) boundary. This enables the maximum possible distance among the actors to be bounded, and also allows them to be close enough to reproduce (as is necessary among most mammals and insects), albeit prey will have to face the dangers of being trapped in an area with predators.

In terms of application, we justify the bounding box conditions by the fact that in nature, ecological resources such as food, water, and living habitats (such as trees and water) can be geographically distant in a spread, and therefore, it is not worth traveling far from one's home for many species (migratory birds and fish are an exception).

However, this restriction can only be applied reasonably to environments of two dimensions and higher, for in the 1D case with bounds of [a, b] and initial coordinates for the prey and predator at x and y (such that $a \leq x, y \leq b$), it is guaranteed that the predator will catch up to the prey after traveling a maximum distance of y-a if $x < y$, or b-y if $y < x$.

Reproductive processes

Reproduction of a species can be viewed as a discrete birth-death process with mixed (jump and stochastic) components. At the beginning of an actor's creation (or birth), it is given zero reproductive energy; this energy level increments at a constant rate (linearly) with respect to time elapsed. Similar to our model of predation, we allow two actors of the same species - i.e. two predators, or two prey - to breed at each unit time if the following tests are satisfied:

1. The two parents have accumulated sufficient reproductive energy above a fixed activation threshold. After reproducing, the reproductive energy is reset to zero. The implication is that there is a minimum delay in between gestation cycles, so that parents cannot procreate non-stop, and newborns must first mature beyond a lower bound of time before they can copulate and contribute to their respective population pool.

2. Let E be an exponentially-distributed random variable with parameter $1/\mathbf{d}(\mathbf{X}_1, \mathbf{X}_2)$ or $1/\mathbf{d}(\mathbf{Y}_1, \mathbf{Y}_2)$, where in this formulation $\mathbf{X}_1$ and $\mathbf{X}_2$ are the two prey, likewise for $\mathbf{Y}$. Then reproduction occurs during one numerical time step if the r.v. E exceeds a certain reproductive rate threshold. For balance to exist in the ecosystem, this reproductive rate threshold must be relatively lower (i.e. reproduction occurs relatively fast enough) for prey compared to that of predators.

Analysis of Experiments

1-dimensional cases for 1 predator and 1 prey

In the simplest case with a single predator and prey, we evaluate the average survival time of the prey and empirical proportion of trials in which the predator manages to catch their prey as a function of these parameters:

i. The capture threshold is the minimum value of the exponential random variable with a rate of 1/d(x,y) that is realized for a predator to catch their prey.

ii. The prey can employ either strategy 0 (pure Brownian motion) or 1 (random walk with drift away from predator). The predator can employ either strategy 0 (pure Brownian motion) or 1 (random walk with drift toward prey).

iii. The predator and prey are initially located independently and identically at position $N(0, \sigma)$.

iv. The arena has a bounding box of [-b, b], where $b > 0$.

We run 10 experimental trials of maximum length 100 steps each per combination of the initial configurations. To compare against the control setting, the baseline case sets a capture threshold of 3, strategy 1 for both predators and prey, normal scale parameter $\sigma=1$, and a [-10, 10] bounding box. Since we only have one predator and one prey in this scenario, no reproduction is possible.

Control: In the base case, it took an average time of T=21.8 steps for the predator to capture the prey.

i. With a lower capture threshold of 1, it took only T=2.7 on average for the predator to catch the prey. When we increase this threshold to 10, the prey survived for 100 time units in half of the trials, and in the remaining n=5 trials, the conditional expected survival time was T=32.2.

In **Figure 2**, we one trial where the prey evaded the predator throughout the entire trial (note how the prey ran from its initial position down to -10, and then up to 10, back and forth, with the predator in pursuit; and on the right, another trial where the predator captured its prey fairly early on.

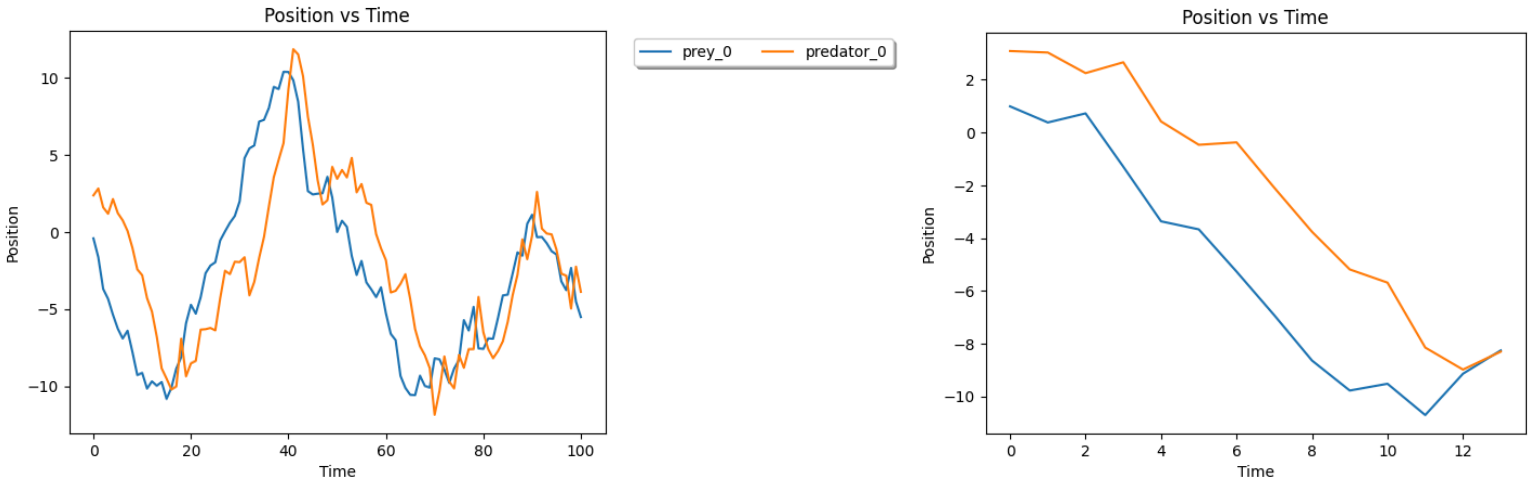

Figure 2. Two example paths for the scenario with one prey and one predator for parameter set (i).

ii. When the predator switches to strategy 0 (pure Brownian motion), it is less efficient at catching prey; in 4 out of 10 trials, the prey survived until the end of the trial, and in the remaining 6 trials, the average survival time for prey was T=40.5.

When the prey alone switches to strategy 0, its chances are grim as the average survival time is only T=11.3; in all cases, it is devoured by the predator.

When both predator and prey switch to strategy 0, the prey survives 2 out of 10 trials, and in the other trials, the average survival time is T=22.75. Below are two sets of sample paths (see **Figure 3**).

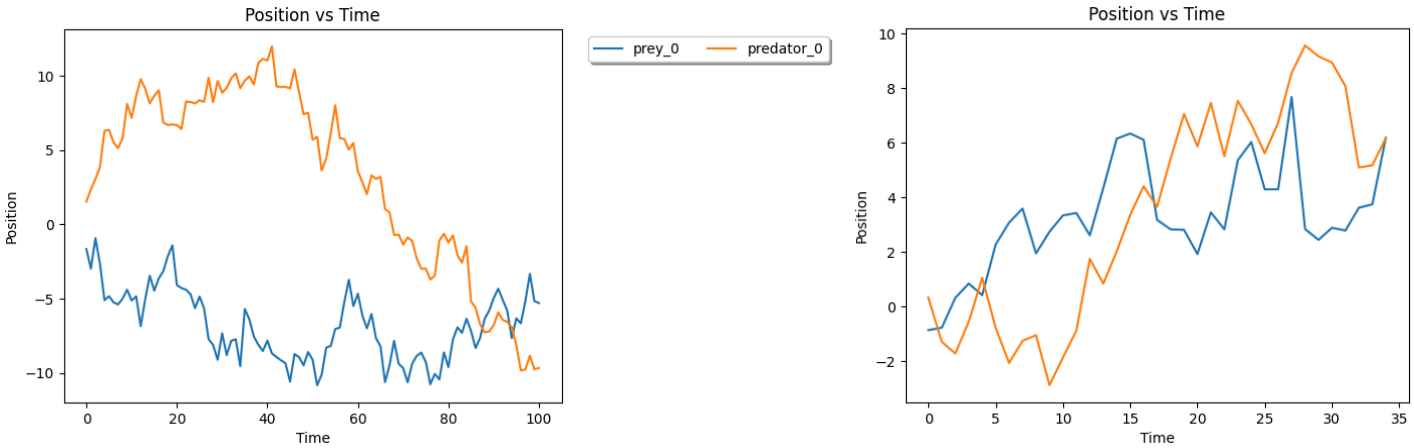

Figure 3. Two example paths for the scenario with one prey and one predator for parameter set (ii).

iii. By reducing the initial dispersion scale to σ=0.5, the predator and prey start off too close to each other, and the prey is always caught, on average T=16.5. By doubling the standard deviation of initial placement to σ=2, the prey's survival times improved to T=26.3 over ten trials (although it was always caught eventually, as it was only a matter of time until the predator catches up in a small enough bounding box).

iv. When we shrink the bounding box to [-1, 1], the prey has a claustrophobic survival time of only T=5.4. But when the bounding box is enlarged to [-100, 100], it escapes in 5 trials and averages T=18 over the other half of the trials. In the plot below, we highlight one trial where the prey continually runs in one direction (with some stochastic noise) and the predator does not manage to catch up.

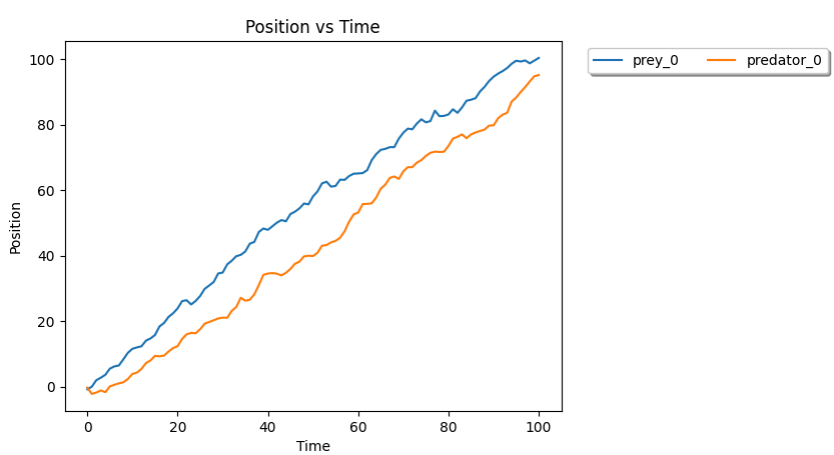

Figure 4. Example path for parameter set (iv) where the prey continually evades the predator

1-dimensional cases for 1 predator and 2 prey

With an additional prey introduced, reproduction becomes a possibility. The influencing factors we investigate are:

v. The minimum reproductive energy M_X must be stored before a prey can reproduce.

vi. The prey have a reproductive threshold value L_X for the simulated exponential random variable and can breed if this value is exceeded.

For the default values, we use $M_X = 5$ and $L_X = 5$. In this base case, reproduction occurred in 2 out of 10 trials, with two prey procreated in one trial (average time of reproduction is $r_X = 36.5$) and five prey procreated in another trial (with $r_X = 38.8$). However, by the end of every trial, the single predator had consumed all existing prey.

In the figures below, we should make it clear that any dead prey who have been killed by a predator will no longer be moving so their positions are fixed (which shows up as a horizontal plateau).

v. When we choose a minimum reproductive energy level of $M_X = 2$, reproduction occurred in two trials as before, but this time we ended up with a total of 268 prey in one trial and 120 prey in the other trial. Only in these two trials did the prey population survive until the very end, and we exhibit the (truncated) plots for these two trials - prey population explodes, so we do not draw the full legend:

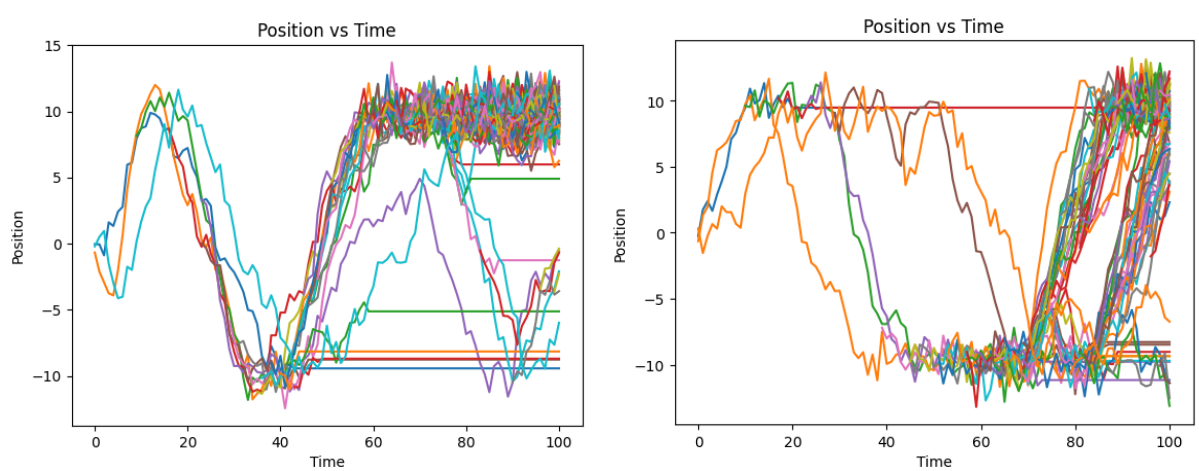

Figure 5. Example paths for parameter set (v)

On the other hand, when we raise the minimum reproductive energy level to $M_X = 10$, the prey populations die out in all trials. However, reproduction occurred in five trials (before the predator killed off the prey parents), and on average 0.6 prey were procreated per trial. Here is one trial where two prey children were born, the parents and one sibling are killed off relatively early, and the remaining child survives until T=98 before it too passes away.

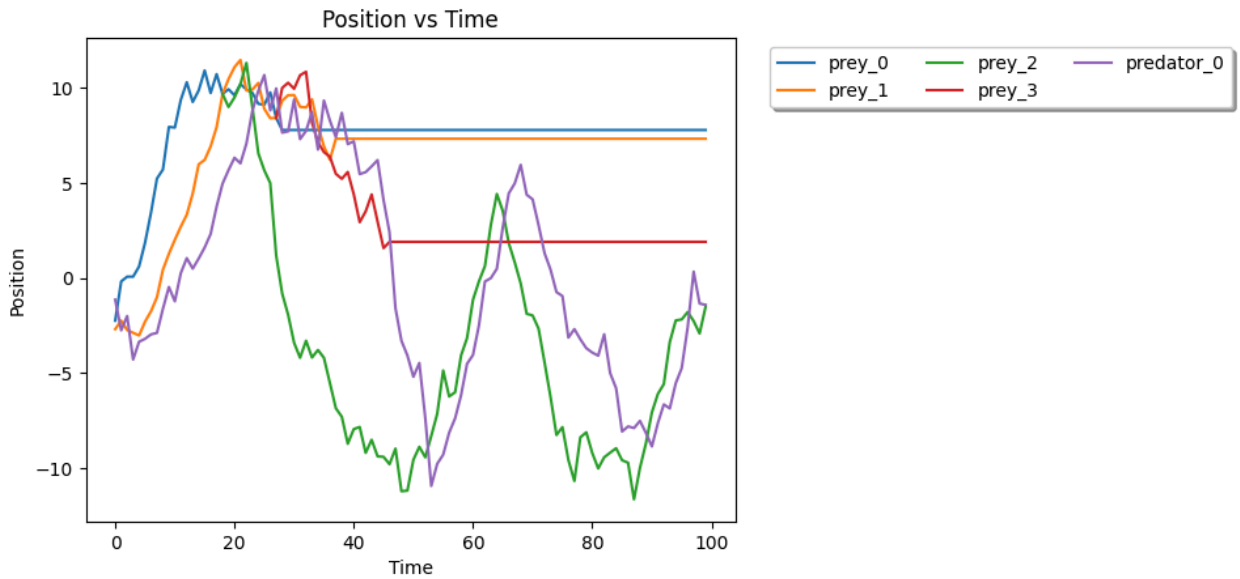

Figure 6. Example paths for parameter set (v) with raised reproductive energy

vi. When $L_X = 2$, reproduction can occur rapidly if the predator does not remove at least one parent of the prey. The average number of prey across the trials is 61. In contrast, when we reduce this setting to $L_X = 10$, the average number of prey across trials is only 2.1, as in all except one trial, the predator ate at least one of the parents before any reproduction could occur.

In summary, the prey's drift function **u** remains unchanged, but in some children's sample paths, the random term involving **f** realizes values that take it farther away from the predator for safety.

1-dimensional cases for 2 predators and 1 prey

For this scenario, we select the default value for predators' minimum reproductive energy as $M_Y = 5$ and the reproductive threshold value of $L_Y = 5$.

In comparison to the case of 1 predator and 1 prey, for the default case here the lone prey survives for T=10 on average, which is lower because now there can be more than one predator chasing their prey.

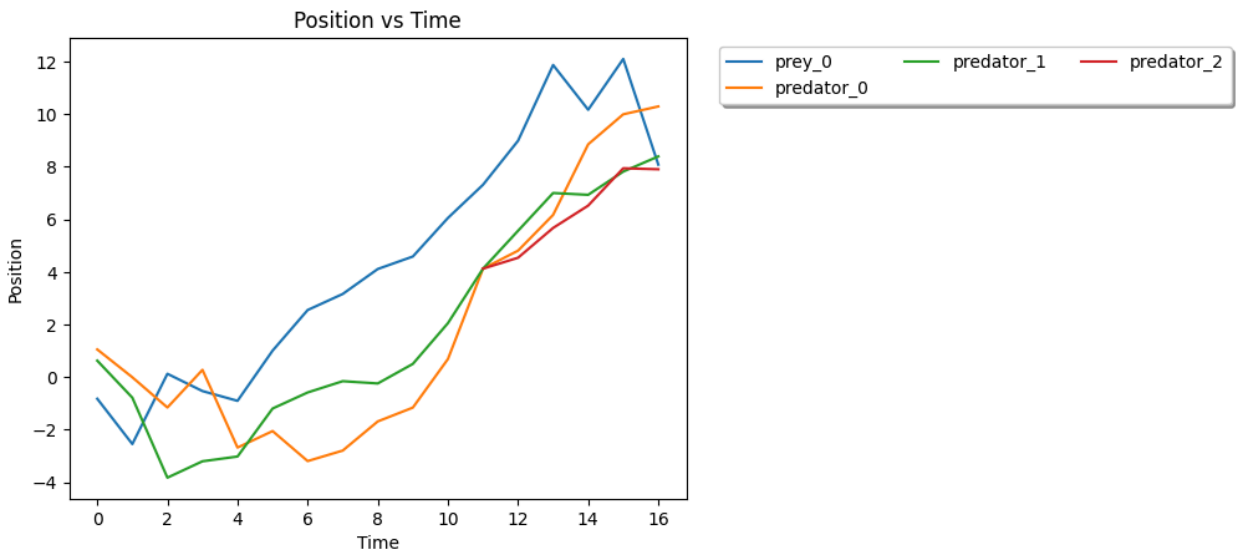

Figure 7. Example paths for two predators and one prey

When the predators' minimum reproductive energy is lowered to $M_Y = 2$, the prey survives on average for T=7.9. When it is raised to $M_Y = 10$, we observed T=9.3 which is not much different because only 0.2 children were born to the predators on average.

When the predators' reproductive threshold is reduced to $L_Y = 2$, the average survival time is T=2.2. At $L_Y = 10$, we found T=6.3, as life is less perilous while your enemies aren't rapidly multiplying to make more hungry carnivores of their kind.

In summary, the predator's drift function **v** is the same as before, but now some predators will draw luckier values from the transformation **g** on the diffusion so that they approach the prey faster than other members of its family.

1-dimensional cases for k predators and n prey

As we have already addressed the effect of each parameter on capture, reproduction, and survival in the previous three sections, we directly report summary statistics for each set of

parameter values on which we experimented. Note that in the following cases onward, we will experiment with 30 trials per run for a larger sample size:

k	n	Average survival time (T)	Percentage of surviving prey	Number of offspring prey	Number of offspring predators
2	2	9.0	0%	0	0
2	5	13.4	0%	6	4
2	10	17.4	23.4%	191	16
5	2	2.9	0%	0	0
5	5	4.2	0%	0	2
5	10	5.2	0%	1	4
10	2	1.9	0%	0	0
10	5	2.3	0%	0	1
10	10	2.7	0%	0	4

Table 1. Results for 1-dimensional cases (30 trials)

2-dimensional cases for 1 predator and 1 prey

For two dimensions, we trace the sample paths from t=0 to t=T rather than plotting over time (because otherwise, we would need animated graphics or 3D plotting, which is not pragmatic for a paper report).

Here are a few examples for the default set of parameter values: capture threshold = 1, prey's strategy = 1, predator's strategy = 1, and bounding box limits = [-10, 10] in each dimension.

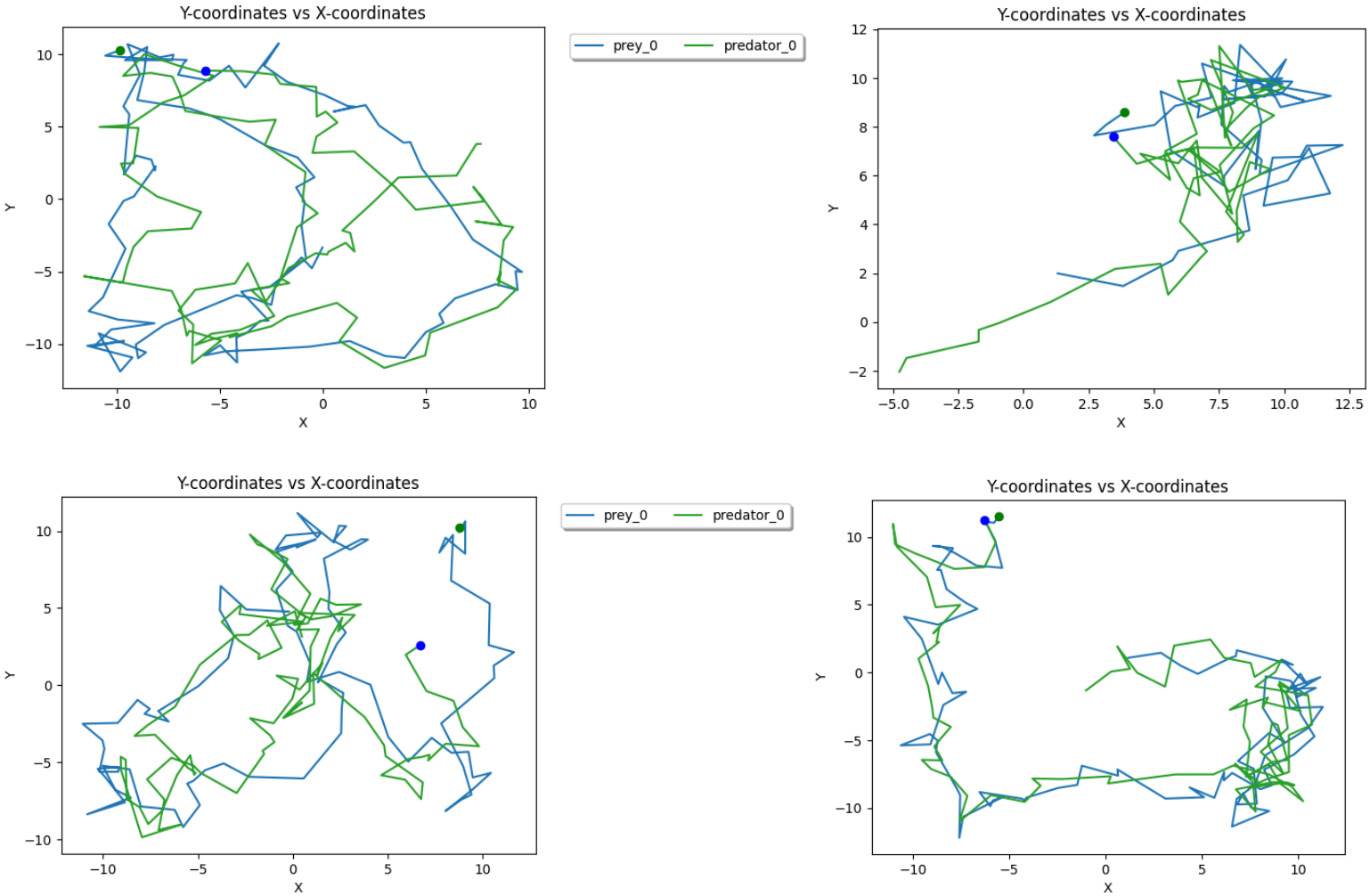

Figure 8. Paths for one predator and one prey in 2D

By varying the parameters, we observe different results according to the summary statistics:

Changed parameter	Average survival time (T)	Percentage of surviving prey
(No change)	72.7	53%
Capture threshold = 1	28.7	3%
Capture threshold = 10	96.0	93%
Prey's strategy = 0	46.6	20%
Predator's strategy = 0	100	100%
Random scatter: $\sigma = 0.5$	69.8	43%

Random scatter: σ = 2	80.7	63%
Bounding box: b = 1	19.6	0%
Bounding box: b = 100	99.6	93%

Table 2. Results for 2-dimensional cases with one prey and one predator (30 trials)

2-dimensional cases for 1 predator and 2 prey

M_X	L_X	Average survival time (T)	Percentage of surviving prey	Average number of offspring prey
2	5	23.2	88%	1.9
10	5	27.1	79%	0.1
5	5	25.9	79%	0.1
5	2	29.0	67%	117.4
5	10	25.6	80%	0.0

Table 3. Results for 2-dimensional cases with one predator and two prey (30 trials)

The default case is highlighted above.

2-dimensional cases for 2 predators and 1 prey

M_Y	L_Y	Average survival time (T)	Percentage of surviving prey
2	5	6.0	73%
10	5	7.0	13%
5	5	7.2	33%
5	2	6.8	63%
5	10	9.0	20%

Table 4. Results for 2-dimensional cases with two predators and one prey (30 trials)

The default case is highlighted above. No children were produced by the pair of predators, as the prey had died too early.

2-dimensional cases for k predators and n prey

All values are default as above, except for the numbers of predators (k) and prey (n) which we vary for collecting observational data..

k	n	Average survival time (T)	Percentage of surviving prey	Average number of offspring prey	Average number of offspring predators
2	2	10.0	72%	0	0.0
2	5	13.8	67%	0.1	0.1
2	10	17.0	54%	28.0	0.6
5	2	3.8	37%	0	0
5	5	2.8	20%	0	0
5	10	6.3	66%	0.0	0.2
10	2	1.6	0%	0	0
10	5	2.1	0.2%	0	0.0
10	10	2.8	0.5%	0	0.2

Table 5. Results for 2-dimensional cases with k predators and n prey (30 trials)

Conclusion

Summary

The results of the experiments in both one-dimension and two-dimension were not surprising. As the reproductive energy of the prey increased, the percentage of the survivors increased; naturally, as the reproductive energy of the predators increased, the percentage of the prey survivors and survival time decreased. We also observed that increasing the size of the arena increased the chance of survival and the survival time for prey. In cases with multiple agents, the most effective strategy for both predator and prey was to pursue or avoid the closest adversary, respectively.

Future Work

As an extension to the simulations we ran in this study, we could also model interactions between different species of predators and different species of prey. For example, different predators may have different affinity for their prey. Additionally, we could model interactions between multiple hierarchies of the food chain, so some predators may have their own predators. As such, we can more closely model ecological communities. In our work, we made some simplifications for ease of analysis; however, we can introduce complex terrains, where the visibility and capture probability of prey and predator would be affected based on where they are in the terrain. Our modeling of reproduction was also simplistic; we could incorporate an intraspecies attraction factor dependent on each species' maturity and reproductive energy, as well as account for the sex of each agent when determining if reproduction was possible.

Works Referenced

Becker, A. T., & Garcia, J. (2018). *The Homicidal Chauffeur Problem*. Wolfram Demonstrations Project.

Huang, L., Chen, X., Tan, X., Chen, X., & Liu, X. (2019). A stochastic predator–prey model for integrated pest management. *Advances in Difference Equations*, 2019(1), 1-10.

Isaacs, R. (1951). Games of pursuit. RAND Corporation.

Gal, S., & Casas, J. (2014). Succession of hide–seek and pursuit–evasion at heterogeneous locations. *Journal of the royal society interface*, 11(94), 20140062.

Merz, A. W. (1971). The Homicidal Chauffeur--a differential game. Stanford University.

Morice, S., Pincebourde, S., Darboux, F., Kaiser, W., & Casas, J. (2013). Predator–prey pursuit–evasion games in structurally complex environments. *Integrative and comparative biology*, 53(5), 767-779.

Patsko, V. S., & Turova, V. L. (2011). Homicidal chauffeur game: History and modern studies. *Advances in Dynamic Games: Theory, Applications, and Numerical Methods for Differential and Stochastic Games*, 227-251.

Tan, X., Qin, W., Tang, G., Xiang, C., & Liu, X. (2019). Models to assess the effects of nonsmooth control and stochastic perturbation on pest control: a pest-natural-enemy ecosystem. *Complexity*.

Weintraub, I. E., Pachter, M., & Garcia, E. (2020, July). An introduction to pursuit-evasion differential games. In 2020 *American Control Conference (ACC)* (pp. 1049-1066). IEEE.

Xiao, D., & Ruan, S. (2001). Global analysis in a predator-prey system with nonmonotonic functional response. *SIAM Journal on Applied Mathematics*, *61*(4), 1445-1472.